Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 · Unit 2 · Introduction to higher order differential equations, complementary function and particular integral

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
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M1

Introduction to higher order differential equations, complementary function and particular integral

1 Introduction to higher order differential equations, complementary function and particular integral

Study focus: Introduction to higher order differential equations, complementary function and particular integral. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I am Gita TR.

Key details

- I will be discussing with you higher order differential equations.
- In this unit, we are discussing about higher order linear differential equation.
- An introduction to higher order linear differential equation.
- Next, we will be learning about homogeneous and non-homogeneous linear differential equation with constant.
- We've been learning methods to solve homogeneous and non-homogeneous LDE linear differential equation with constant coefficients.
- We are also discussing about linear differential equation of higher order with variance.
- In fact, we are discussing two different forms of linear differential equation with variable coefficients and methods to solve the two different forms of linear differential equation with variable coefficients.
- We are also learning a method called method of variation of parameters, which is used to solve only second order linear differential equation.
- And we are learning at the end application of higher order linear differential equation to solve problems on electric circuits and simple harmonic motion.
- First is the definition of linear differential equation.
- We are also learning the general form of higher order linear differential equation.
- We are learning different forms of higher order linear differential equation.

- Linear differential equation with constant and with variable coefficients.
- We are learning a very important concept called complementary function and particular integration.
- Why do we have to study differential equations?
- We know that differential equation play a very prominent role in the field of engineering, science, and economics and other disciplines.
- They serve as mathematical models to solve any real world problem.
- They help in modeling every physical, biological and technical process.
- Whether the order of the differential equation is 1 or higher than 1, more than 1, ordinary differential equation is classified into two types based on linearity.
- Non-linear differential equation.
- Is a linear differential equation.
- A differential equation which is not linear is a nonlinear differential equation.
- that this both the equations are of order 1, this is LDE of order 1, and this is nonlinear differential equation of order 1.
- And in the previous sessions, you learned methods to solve both LDE of order 1 and non-linear differential equation of order 1.
- But in the coming session,.
- Sessions we will be talking about only linear differential equation.
- What is x here? x is the independent variable and y the dependent variable which is dependent on x.
- Do you call this differential equation a linear differential equation?

Example or application from the lecture

- Let us first look at the content of this unit.
- Let us just write an example for linear and nonlinear differential equation.
- An example for linear differential equation is dyed by dx plus 2x into y is equal to x square and an example for nonlinear differential equation is y square d by dx plus sine y is equal to x square e to the power 2y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

and its derivative occur in first degree.

Key details

- And are not multiplied together is a linear differential equation.
- Look at this differential equation.
- It is y to the power 1, dy by dx to the power 1.
- Also, the equation do not contain the dependent variable and its derivative in product form.
- Do not have y into dy by dx.
- What is the odd of the differential equation.

- Order of a differential equation is the order of the highest order derivative present in the differential equation.
- It is dy by dx.
- But in the sessions we will be talking about only higher order linear differential equations.
- Definition goes this way.
- A differential equation is said to be linear if the coefficient y and its derivatives are functions of x are constants.
- The degree of y and its derivatives is 1.
- Now, first order linear differential equation.
- We know it is of the form dy by dx plus p into y is equal to q, where p and q are functions of x alone.
- See how does it look like a different equation the form P naught into D2y by Dx2 plus P1 into Dy by Dx plus P2 into Y is equal to capital X where.
- P0 P1 P2 and capital X are functions of the independent variable X and P0 equal to 0 is a linear differential equation of Albert 2.
- Why it is a linear different differential equation?
- What is the definition of linear differential equation?
- A differential equation in which two conditions.
- What is the first condition?
- The coefficient of y and.
- Its derivatives should be functions of x.
- The coefficient of y is p2, dy by dx is p1, and p naught, they are functions of x.
- And also the degree of y and all its derivatives is 1 y to the power 1 d y by dx to the power.
- 1 d2 y by d is 2 to the pi 1.
- Because the order of this differential equation is 2, so it is a second order linear differential equation.
- P0 P1 P2 P n or functions of X alone is a linear differential equation of R n provided P naught not equal to 0.
- The coefficients P naught P1 P2 Pn are functions of x.
- And also the co and also the degree of y.
- It's it all its derivatives, it is dn y by dx n to the power 1.
- n minus one third or derivative of y with respect to x to the power one dn minus two y by dx n minus two to the power one.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

Plus E a n into Y is equal to capital X where E naught even into A N are either functions of X or constants.

Key details

- capital X a function of X.
- What are the two different forms?

- One is linear differential equation with constant coefficients.
- Look at the coefficients of the derivatives.
- It is the coefficient of y is 1, dy by dx is 7, d2 y by dx2 is minus p, d3 y by dx3 is 2.
- What is the order of.
- The differential equation.
- Then the differential equation is linear differential equation with variable coefficient a01 a2 a n are functions of x.
- 1 can be written as 1 into x to the power 0.
- Methods to solve linear differential equation with constant coefficients.
- LDE with constant coefficients can be classified into two types based on this cap right hand side function capital X.
- One is called homogeneous linear differential equation.
- The linear differential equation with constant coefficients in which.
- This capital X right hand side function is zero.
- And the other one is non-homogeneous differential equation.
- If RHS function is a non-zero function of the independent variable, then the differential equation is non-homogeneous LD.
- RHS capital X is 0, RHS capital X is a 10 function of X.
- Equal to 0.
- Because RHS is 0, it is a homogeneous LDE.
- See, I have taken the same.
- Every non-homogeneous linear differential.
- Equation has a homogeneous linear differential equation associated with it.

Example or application from the lecture

- So an example for it, 2d3 by dx3 minus 5, d2y by dx2 plus 7 dy by dx plus y is equal to x square.
- An example for this is 2x cube into d3y by dx3 minus 5x square plus 2 into d2 y by dx2 plus x into dy by dx plus y is equal to x square into sine x.
- So it is an example for linear differential equation of order 3 and with variable coefficients.

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EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

Let us learn about the differential operator.

Key details

- A differential equation.
- What is this differential equation?
- A1, a2, a3, a and say they are constants.
- And linear differential equation with constant coefficient, and it is a non-homogeneous linear differential equation.

- plus a1 d to the power n minus 1 of y plus a2 d to the power n minus 2 of y plus a n into y is equal to capital X.
- You can factorize this f into linear factors, and the linear factors can be written in any order.
- This can be treated like an algebraic expression and it obeys the laws of algebra.
- The linear differential operator.
- The first component is called complementary function, and the second component is called particular invariant.
- And this complementary function is the complete solution of the associated homogeneous linear differential equation.
- That the solution of the whole non-homogeneous linear differential equation has two components.
- One is called complementary function, and the other one is called particular integral.
- And asked to solve the non-homogeneous linear differential equation, you need to find the complementary function and the particular integral of the differential equation.
- and dependence is defined for n number of functions.
- A set of functions f1, f2, f3, fn are said to be linearly dependent on an interval i.
- If there exists constant c1 and c2, c three, cn, not all zeros such that c1 into f1 plus c2 into f2 plus c n into f n is equal to 0.
- What do you mean by this?
- See the linear combination of the functions f1 f2 f n which is c1 f n plus c2 f2 plus c n f n is equal to 0 even for non-zero constant c1 c2 c three c n then the function cf1 f2 f n are said to be linearly dependent functions.
- What are linearly independent?
- If not linearly dependent, the functions are called linearly independent functions.
- The set of linearly independent.
- If the set is linearly independent, so what do you mean by it?

Example or application from the lecture

- We know that the first order differential operator is given by d by ts.
- capital D is the notation given for the first order differential operator D by Dx.
- Now let us first discuss in detail about this complementary function.

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EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

equal to 0 only when c1 c2 c 3 cn 0.

Key details

- That is linearly independent function.
- The linear combination of the function c 1 f1 plus c2 f2 plus c n f n is equal to 0.
- It is true only when the arbitrary constant c1 c2 c3 c n are zeros on that is a trivial case otherwise this is not equal to zero for non-zero arbitrary constants c one c2 c three c n this is not equal to zero if f1 f2 f3 fn are leaving.

- Linearly independent of each other.
- See two functions y now y1 of x and y2 of x are said to be linearly independent when the linear combination c1 into y1 plus c2 into y2 is not equal to 0.
- 0 for non-zero constants c1 and c2.
- That is C C1 by 1 is not equal to minus C2 into Y2.
- That means that y1 not equal to minus C2 by C1 into.
- R y1 is it not equal to minus c2 by c1 into y2.
- Two functions to be linearly independent of each other.
- Neither of the function is a constant multiple of the other.
- That is, y1 cannot be written as some constant into y2, or y2 cannot be written as some constant into y1.
- Then the functions y1 and y2 are said to.
- Be linearly independent functions.
- Like see, this is homogeneous because RHS is 0, K1, K2 are constant, so it is a linear differential equation with constant coefficient.
- How does the general solution of this differential equation?
- See, now it's very important if y1 and y2 are two linear.
- linearly independent solutions of this.
- Second order homogeneous linear differential equation with constant coefficients, then the linear combination C1 by 1 plus C2Y2 is also a solution of the differential equation.
- See, please understand if y1 and y is 2 are linearly independent, then the linear combination C1 by 1 plus C2Y2 is also a solution.
- Why C1 by 1 plus C2Y2 is a solution only when y1 and y2 are linearly independent.
- independent functions, but if y 1 and y2 are linearly independent, then the linear combination c1 by 1 plus c2 by 2 is not equal to 0 for non-zero constant c1 and c2.
- And this is also a solution of the differential equation.
- In fact, this represents the complete solution of the differential equation.
- Very important concept.
- The complete solution of homogeneous linear.
- differential equation with constant coefficient is of the type c1 y 1 plus c2 y2 where.
- Y1 and Y2 are two linearly independent functions.
- n third d n y by dxn plus a1 into n minus one third derivative of y with respect to a x plus a n of y is equal to 0 where a1 a2 a3 a n are constant.

Example or application from the lecture

- Linear differential equation with constant coefficients by first taking an example of now.
- Let us consider a second order LDE with constant coefficients in the type d2 y by dx2 plus k1 dy by dx plus k2 div is equal to 0.

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EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 25:00–29:59

has n linearly independent solutions y1 by two y three y ninear combinations c1 by one plus c2 by two plus cn by n also represent a solution of the same differential equation and it is the complete solution of the differential equation.

Key details

- And the complete solution of the differential equation is called complementary function of the differential equation.
- The complete solution of linear differential equation, which is homogeneous, and it is of order n will be of the form.
- is equal to c1 by 1 plus c2 by 2 plus cn by n and the complete solution is called complementary function of the differential equation.
- With constant coefficients is called complementary function of the differential equations.
- F of d of y is equal to capital X, where f of d is the linear differential operator.
- One is the complementary solution, other one is a particular solution.
- What do we mean by this first statement?
- Homogeneous linear differential equation.
- The associated homogeneous differential equation is f of d of y is equal to 0.
- Y is equal to YP is a particular solution.
- One solution of the non-homogeneous differential equation f of d of y is equal to capital S, then the complete solution of this differential equation is of the form y is equal to yc.
- C plus YP.
- Why it is called as the particular solution about which we will be learning later in a different session.
- What is Yc?
- It is the complete solution of the corresponding homogeneous differential.
- This particular solution YP is also called particular integral.
- Now in the next session, we will be learning.
- Methods to find the complementary function of the differential equation.

Example or application from the lecture

- C f of d of y is equal to capital X is the given non-homo.

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EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Introduction to higher order differential equations, complementary function and particular integral transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to higher order differential equations, complementary function and particular integral transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to higher order differential equations, complementary function and particular integral transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.
Introduction to higher order differential equations, complementary function and particular integral transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/001_Introduction_to_higher_order_differential_equations_complementary_function_and_particular_integral__069274588a013075.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.

2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00002	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00003	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00004	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00005	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

C Revision and retrieval practice